Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam B · Form 2: Pulley Focus

Multiple Choice Questions: Select the best answer. (1 point each)

1. A motion sensor sees the cart on a smooth table speed up toward an ideal pulley. What must be true of the hanging mass while the rope stays taut?

(A) It has zero acceleration because tension is uniform

(B) Its downward net force is positive

(C) Its tension exceeds its weight

(D) It travels only half the cart distance

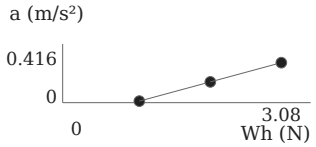
2. A horizontal cart slides toward an ideal pulley. Cart plus hanger mass is fixed at 5 kg; a constant 1 N resistance acts on the cart. Use g=10 m/s². Wh (N) -> a (m/s²): (1.08, 0.016); (2.08, 0.216); (3.08, 0.416). What is the slope between the endpoint measurements?

(A) 0.12

(B) 0.28

(C) 0.2

(D) 0.36



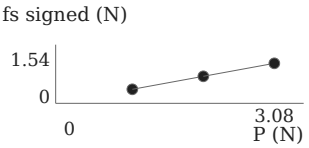
3. An incline block of weight 10 N rests on a 30° ramp with a 5 N hanger on an ideal pulley. Add vertical downward force P to the block. Friction is sufficient; uphill is positive. P (N) -> fs signed (N): (1.08, 0.54); (2.08, 1.04); (3.08, 1.54). Which relation models this apparatus?

(A) Y=-0.5X+(0)

(B) Y=1X+(0)

(C) Y=0.5X²+(0)

(D) fs=(W+P)sinθ-Wh



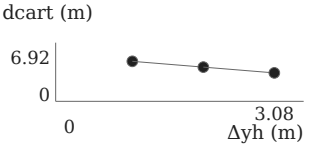
4. A cart and a hanging mass share a taut rope over one ideal fixed pulley. Measure downward hanger displacement and the cart distance from the pulley. Delta yh (m) -> dcart (m): (1.08, 6.92); (2.08, 5.92); (3.08, 4.92). Predict the reading at horizontal coordinate 2.58.

(A) 3.252

(B) 7.588

(C) 9.756

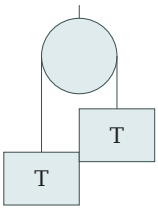
(D) 5.42



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. Two vertical rope strands pass around a weightless fixed pulley. Tension in each is 18 N.

a. Draw both forces exerted by the rope on the pulley alone. **(1 pt)**



b. Find the support force and explain why the two strand forces are not a third-law pair. **(3 pt)**

6. A stationary incline block with 3 N of downhill weight is connected to a hanger over an ideal pulley. Static friction is sufficient; count uphill friction positive. Wh (N) -> fs signed (N): (1.08, 1.92); (2.08, 0.92); (3.08, -0.08).

a. Build the model for static friction on a ramp from physics and predict both slope and intercept. **(2 pt)**

b. Label the axes and units and demonstrate interpolation between two readings. **(2 pt)**

7. A stationary weight W hangs from each side of a weightless fixed pulley. Both rope segments are vertical. Derive the upward support force and distinguish it from the tension. **(3 pt)**